Repair Bias in Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Repair Bias in Data

> Prepare Repair Bias in Data: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Identify selection and coverage **bias**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Sampling and Representation

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Define a question and target population

Reference example, not the only acceptable wording: Population: all Grade 7 students. Only lunch-club members or one friend group is an unsuitable basis for a whole-grade claim.

Noise and Outliers

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate mean and median

Reference example, not the only acceptable wording: Sum 30 divided by 5 gives 6.

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

Concept to explain

Fairness work changes data, features, tests, or use conditions, then rechecks outcomes.

Term	Meaning
bias	A systematic tendency in data, decisions or outcomes that can distort a result or disadvantage people.
rebalance	Change the relative representation of groups or classes in data using a stated method.
subgroup	A defined subset of cases considered separately in analysis.
retest	Run a test again after a change or under specified new conditions.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 20 as a two-session investigation using fictional data. A fictional poll has 8 Yes among 10 club respondents and 2 Yes among 10 other respondents.

If only club members are surveyed, an 80 percent result misses a group with a different pattern. Examine who is included before treating a larger sample as better.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

The population is the full target group; the sample is the observed subset.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can distinguish a **bias**ed collection **process** from a mere small dataset.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Repair Bias in Data using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Audit: quiet-room voices work; noisy-room voices often fail.

2. Possible causes: missing environments, microphone differences, narrow speakers.

3. Repair: collect consented examples across settings; remove shortcut features.

4. **Retest** the same **subgroup** cases and report before/after results.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Training has eighteen adventure readers and two poetry readers; many intended users

prefer poetry.

The teaching case

Propose repair and evaluation.

Exact explanation to model

Why the result follows

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

Explain the added worked example

A fictional poll has 8 Yes among 10 club respondents and 2 Yes among 10 other respondents. If only club members are surveyed, an 80 percent result misses a group with a different pattern.

Examine who is included before treating a larger sample as better.

The combined observed sample has 50 percent Yes. This is not automatically a population estimate: we also need to know how the groups were selected and how many people each represents.

Do not average percentages without considering their denominators.

Key terms

Term	Meaning
Selection bias	Systematic distortion caused by who enters the data
Undercoverage	Leaving part of the target group out of selection
Nonresponse	A selected participant does not respond
Subgroup	A defined part of a larger group

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Suppose 90 club respondents give 72 Yes and 10 others give 2 Yes. What is the combined observed proportion?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Repair Bias in Data

A fictional school polls learners about adding comic recommendations.

The facts we will use

> **Supplied facts**
>
> Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15.

Supplied facts

The full grade has 30 club members and 90 others. These respondents were volunteers, not random selections.

AI ACADEMY / AI-2/20

Volunteer group comparison

	Yes responses	Respondents	Grade members
Club	12	15	30
Others	3	15	90

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/20

Repair Bias in Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group rates

Club:12/15=80%; others:3/15=20%.



2. Respondent total

15 Yes out of 30 respondents =50%.



3. Check representation

Respondents split 50/50, but the grade is 25% club and 75% other.



4. Repair collection

Plan including equality, documented invitations and report nonresponse by group.

CHANGE ONE THING

Recruit more club volunteers only.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Group rates	Club:12/15=80%; others:3/15=20%.
2. Respondent total	15 Yes out of 30 respondents =50%.
3. Check representation	Respondents split 50/50, but the grade is 25% club and 75% other.
4. Repair collection	Plan including equality, documented invitations and report nonresponse by group.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact.

What this example does not establish

Keep the stated limit

Neither raw nor weighted volunteer rates prove the whole grade's opinion; privacy and consent rules still apply.

Change one thing in the pictured case

Change one thing

Change: Recruit more club volunteers only.

Predicted result

Prediction: Sample size increases without fixing missing representation of other learners.

Why the result follows

Reason: The added respondents come from the already overrepresented group.

Ask while showing the picture

Ask while showing the picture · 1

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Repair Bias in Data, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: **Training** has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

2. State the decision to explain

Write the question to explain: Propose repair and evaluation.

3. Reveal the worked reasoning

Reveal after **prediction**s: Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

Hold back the independent task

Keep for an independent attempt: A dataset has 18 adventure readers and 2 poetry readers. A recommender ignores poetry preferences. What would you inspect and test after a repair?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

74/100 = 74 percent. The sample is dominated by club respondents. Exit: No. Existing interest could influence club membership, and other factors may differ.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage	Teacher action
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Group A has 3 Yes of 4 and B has 2 Yes of 6. Find the combined proportion.

Reasoning to listen for

Teacher check: 5/10 = 50 percent. The unweighted average of 75 percent and about 33.3 percent is not the combined sample rate.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Selection bias** and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Group differences suggest that coverage matters, but do not explain why choices differ. Small group counts also make percentages unstable.

Use neutral group labels in fictional data rather than personal or sensitive attributes.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

Reasoned teacher answer

It assumes each volunteer group rate represents its whole group; that assumption has not been established.

Changed-case reasoning: Sample size increases without fixing missing representation of other learners. The added respondents come from the already overrepresented group.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Repair Bias in Data investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Given an uneven dataset, name the missing coverage, choose one repair, predict an outcome, and rerun a fixed **subgroup** test set.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: Group differences suggest that coverage matters, but do not explain why choices differ. Small group counts also make percentages unstable.

Use neutral group labels in fictional data rather than personal or sensitive attributes.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: A 6/8 and B 1/2 combine to 7/10 = 70 percent, with group rates 75 and 50 percent. Require a selection limitation.

A convenient shortcut preserves meaning: Surveying another 100 people from the same undercovered source may make the sample larger without fixing the selection problem.

Improve access to missing groups and record **nonresponse**; do not simply duplicate existing rows.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Repair Bias in Data questions, teacher keys and criteria after the relevant attempts.

> Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

Independent question

Independent question · withhold the key

A dataset has 18 adventure readers and 2 poetry readers. A recommender ignores poetry preferences. What would you inspect and test after a repair?

Expected independent answer

Teacher key · Expected independent answer

Inspect representation and recommendation results for poetry readers. Gather relevant varied examples and compare performance for both interests on separate test cases.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Duplicating one poetry record supplies diverse **evidence**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Add varied relevant poetry titles with consistent metadata, keep scoring fixed and compare results on the same validation requests by interest. Then inspect whether scoring still disadvantages poetry.

Save fresh final requests and do not infer cause from suggestion counts alone.

Explanations for the additional workbook examples

W1 Locate undercoverage

Workbook question · W1

A morning-only survey claims to represent all library visitors. Which visitors may be missed?

Teacher reasoning · W1

Afternoon or evening visitors, and people unable to attend in the morning.

W2 Combine counts

Workbook question · W2

Group A has 3 Yes of 4 and B has 2 Yes of 6. Find the combined proportion.

Teacher reasoning · W2

5/10 = 50 percent. The unweighted average of 75 percent and about 33.3 percent is not the combined sample rate.

W3 Repair the collection

Workbook question · W3

A form is shared only in the robotics club. How can the **process** better cover the grade?

Teacher reasoning · W3

Use a full-grade selection frame and access channels available across groups, with a **nonresponse** record. State remaining limitations.

W4 Audit independently

Workbook question · W4

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

Teacher reasoning · W4

90 percent, 20 percent and 20/30 about 66.7 percent. Group sampling proportions and response patterns may differ from the grade.

W5 Avoid causal claims

Workbook question · W5

Club members favor coding more often. Does club membership necessarily cause that preference?

Teacher reasoning · W5

No. Existing interest could influence club membership, and other factors may differ.

Assess independent transfer

Independent question · withhold the key

Selected case: A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

Selected case answer

Teacher key · Selected case answer

90 percent, 20 percent and 20/30 about 66.7 percent. Group sampling proportions and response patterns may differ from the grade.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Repair Bias in Data to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Change feature coverage or weights only after locating which reader profiles are underserved.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 21: Similarity by Selected Traits. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use two colors of anonymous group cards. Extension: compare combined rates after changing only sample group sizes while preserving group rates.

Independent W4 **evidence**: 90 percent, 20 percent and 20/30 about 66.7 percent. Group sampling proportions and response patterns may differ from the grade. Record whether the learner needed no help, a hint or explanation.

Fresh retry: A 6/8 and B 1/2 combine to 7/10 = 70 percent, with group rates 75 and 50 percent. Require a selection limitation.

Next week: Relationships and confounding. Reuse this week's explicit contracts and **evidence**.

Plan the delayed return

Suggested return points in this level: ai-2/21, ai-2/23, ai-2/26. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Repair Bias in Data lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Identify selection and coverage **bias**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Group A has 3 Yes of 4 and B has 2 Yes of 6. Find the combined proportion.

> ### Supported practice answer
>
> Reasoned practice answer: 5/10 = 50 percent. The unweighted average of 75 percent and about 33.3 percent is not the combined sample rate.

Check every learner before discussion: Use the worked case for **Selection bias** and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Group differences suggest that coverage matters, but do not explain why choices differ. Small group counts also make percentages unstable.

Use neutral group labels in fictional data rather than personal or sensitive attributes.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Recruit more club volunteers only. Require a **prediction** and a reason before comparing. Expected result: Sample size increases without fixing missing representation of other learners.

Reason: The added respondents come from the already overrepresented group.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately

Criterion	Not yet	With support	Independent
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

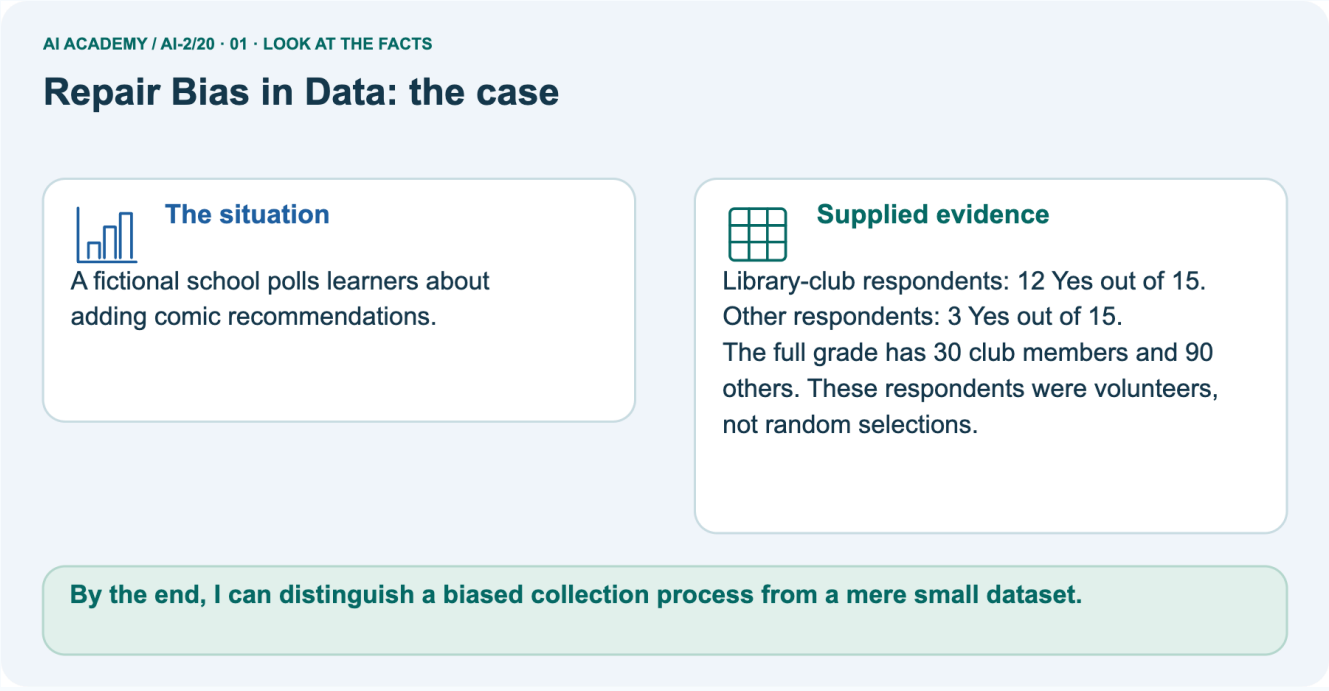
TEACHER GUIDE · 9 OF 9

Visual concept journey

> Show one Repair Bias in Data graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Repair Bias in Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional school polls learners about adding comic recommendations. Supplied facts: Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15. The full grade has 30 club members and 90 others.

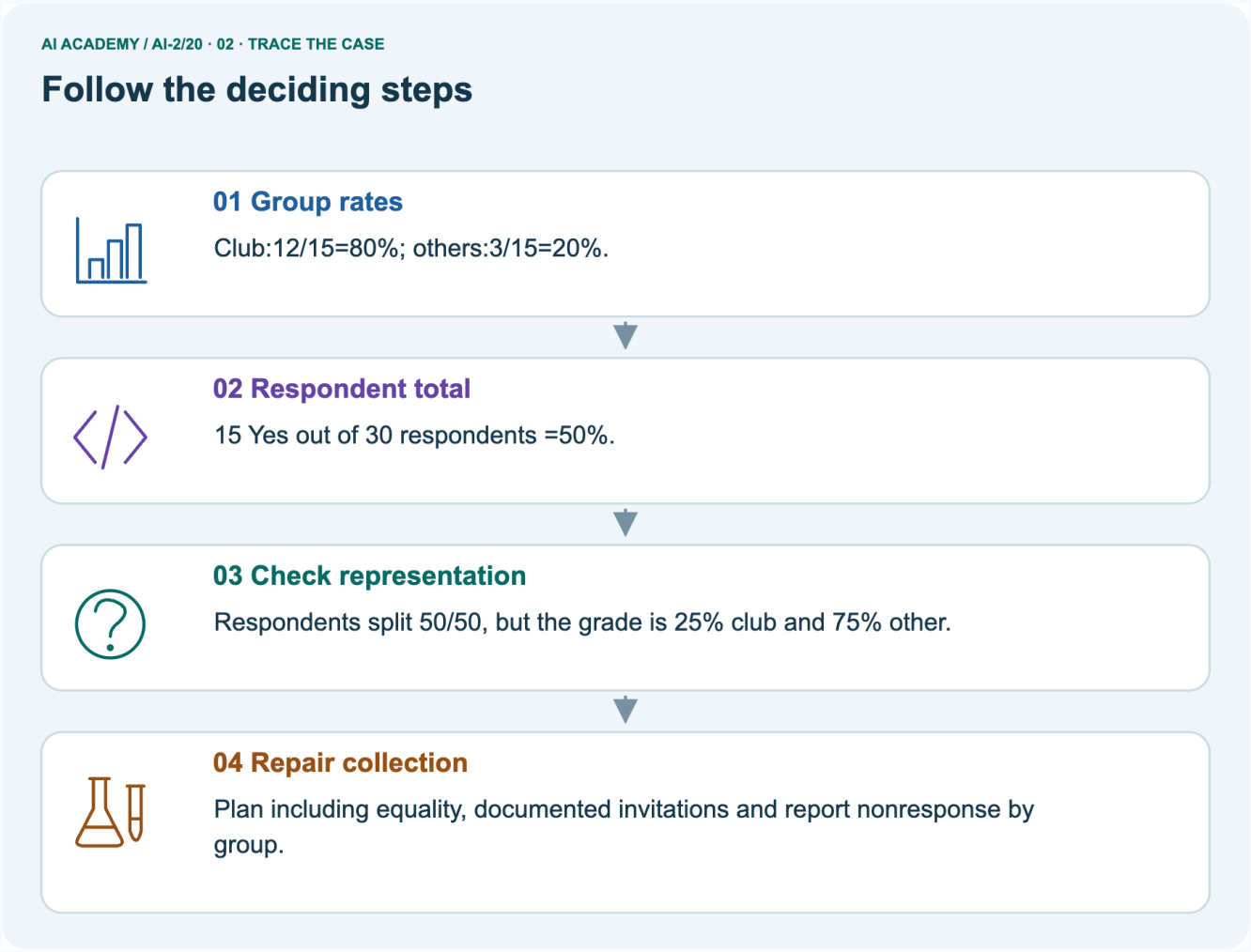
These respondents were volunteers, not random selections.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

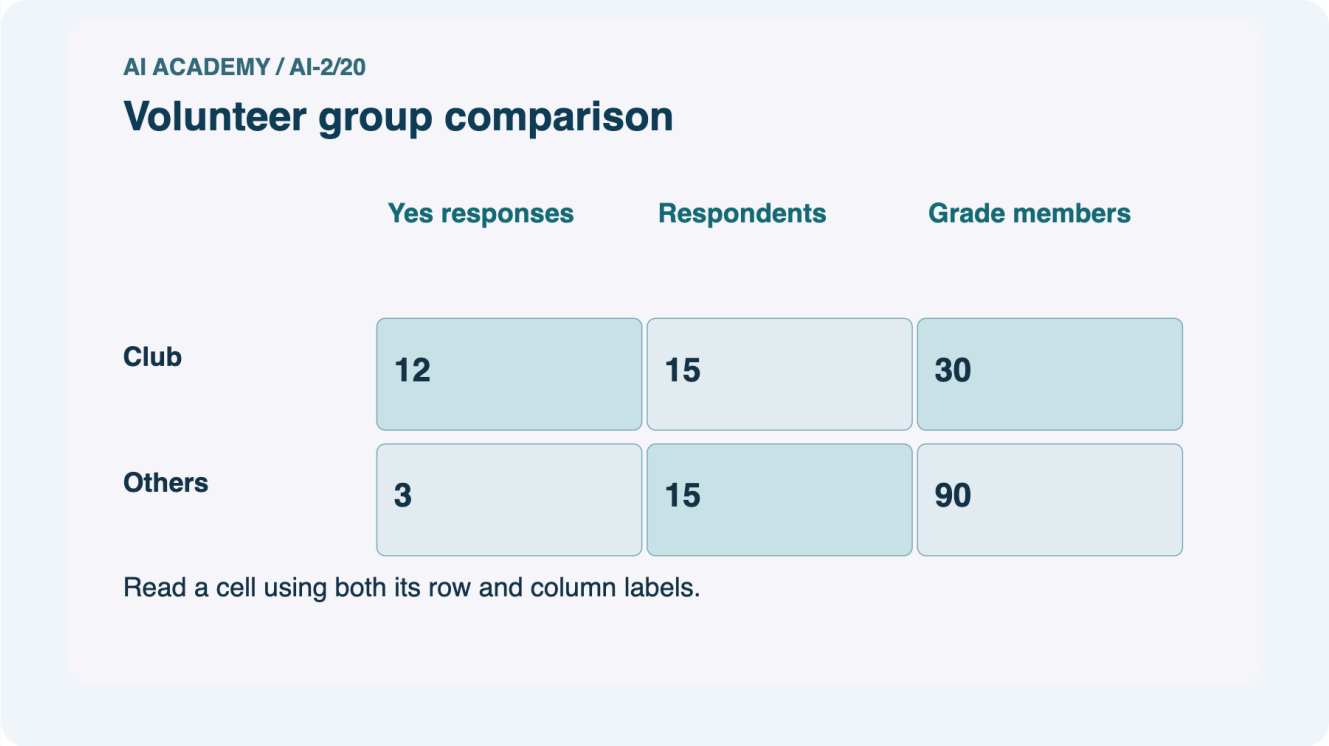
Group rates: Club:12/15=**80%**; others:3/15=20%. → Respondent total: 15 Yes out of 30 respondents =50%. → Check representation: Respondents split 50/50, but the grade is 25% club and 75% other. → Repair collection: Plan including equality, documented invitations and report **nonresponse** by group.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Volunteer group comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Volunteer group comparison. Club: Yes responses = 12, Respondents = 15, Grade members = 30. Others: Yes responses = 3, Respondents = 15, Grade members = 90.

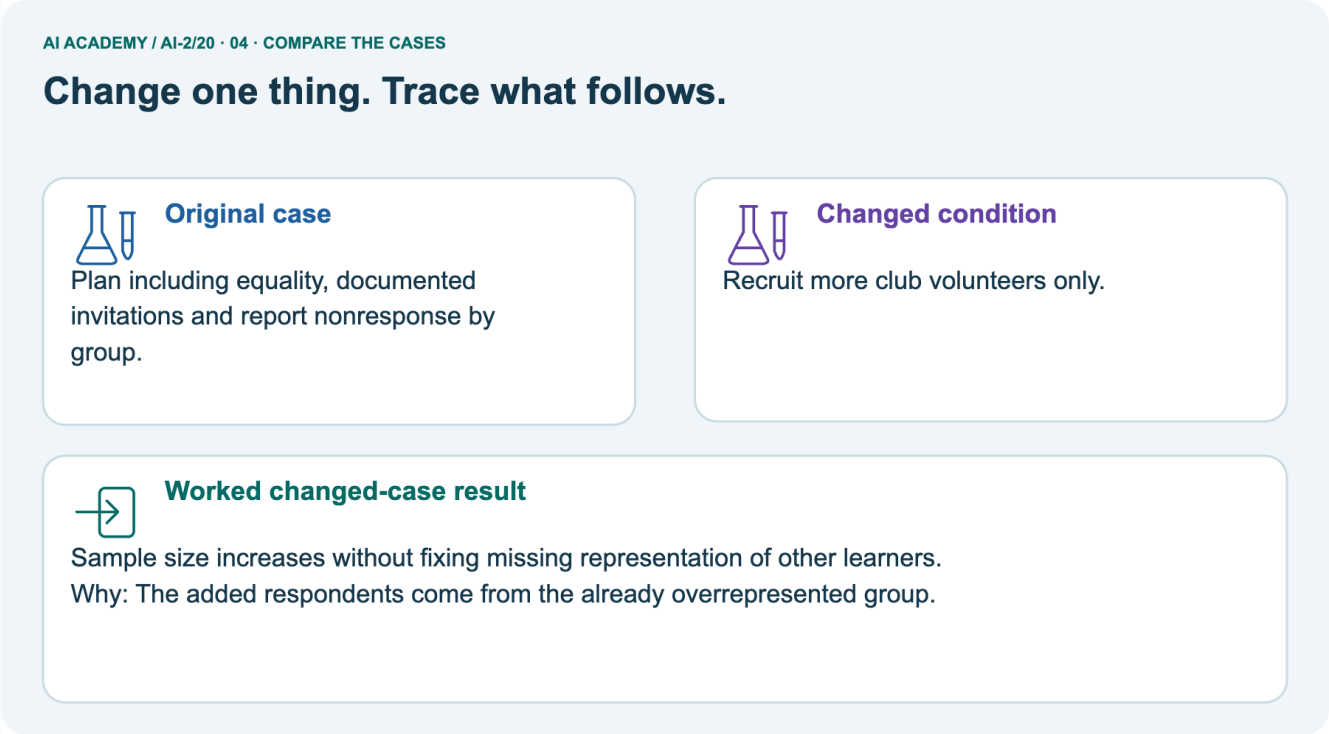
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Plan including equality, documented invitations and report **nonresponse** by group.
Changed condition: Recruit more club volunteers only.

Expected result: Sample size increases without fixing missing representation of other learners.
Reason: The added respondents come from the already overrepresented group.

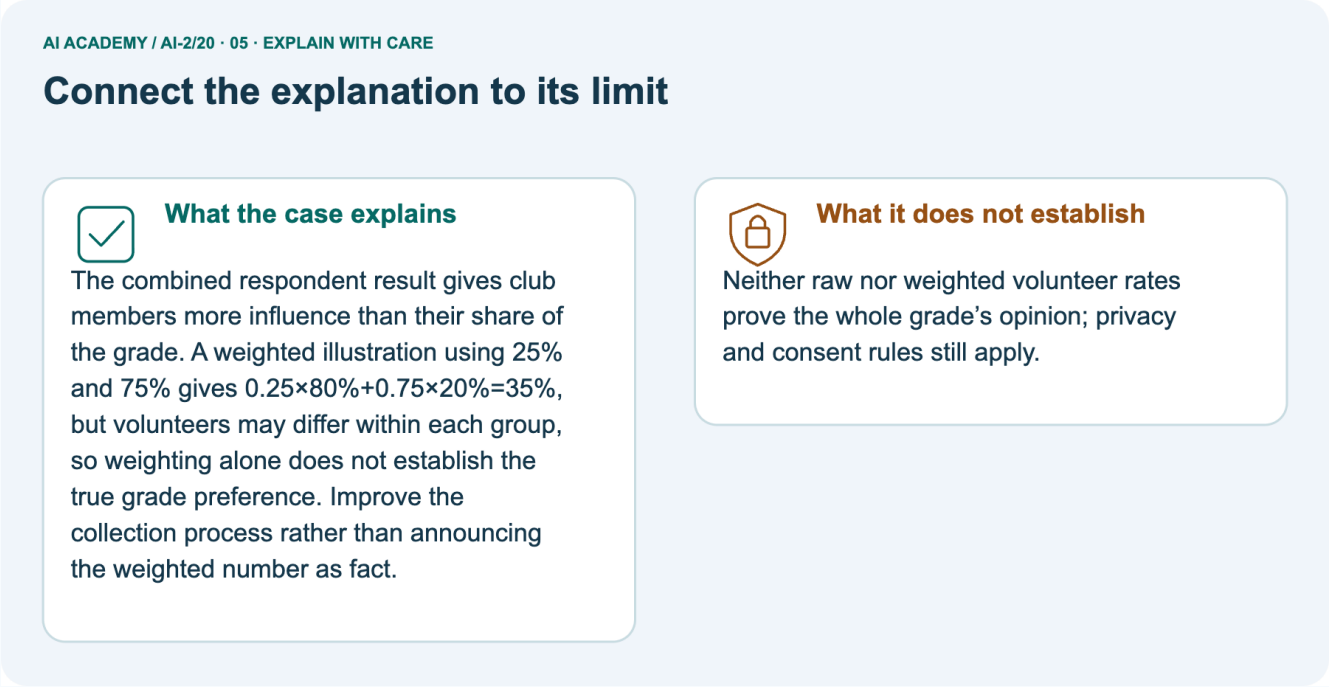
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The added respondents come from the already overrepresented group.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact. Limit: Neither raw nor weighted volunteer rates prove the whole grade’s opinion; privacy and consent rules still apply.

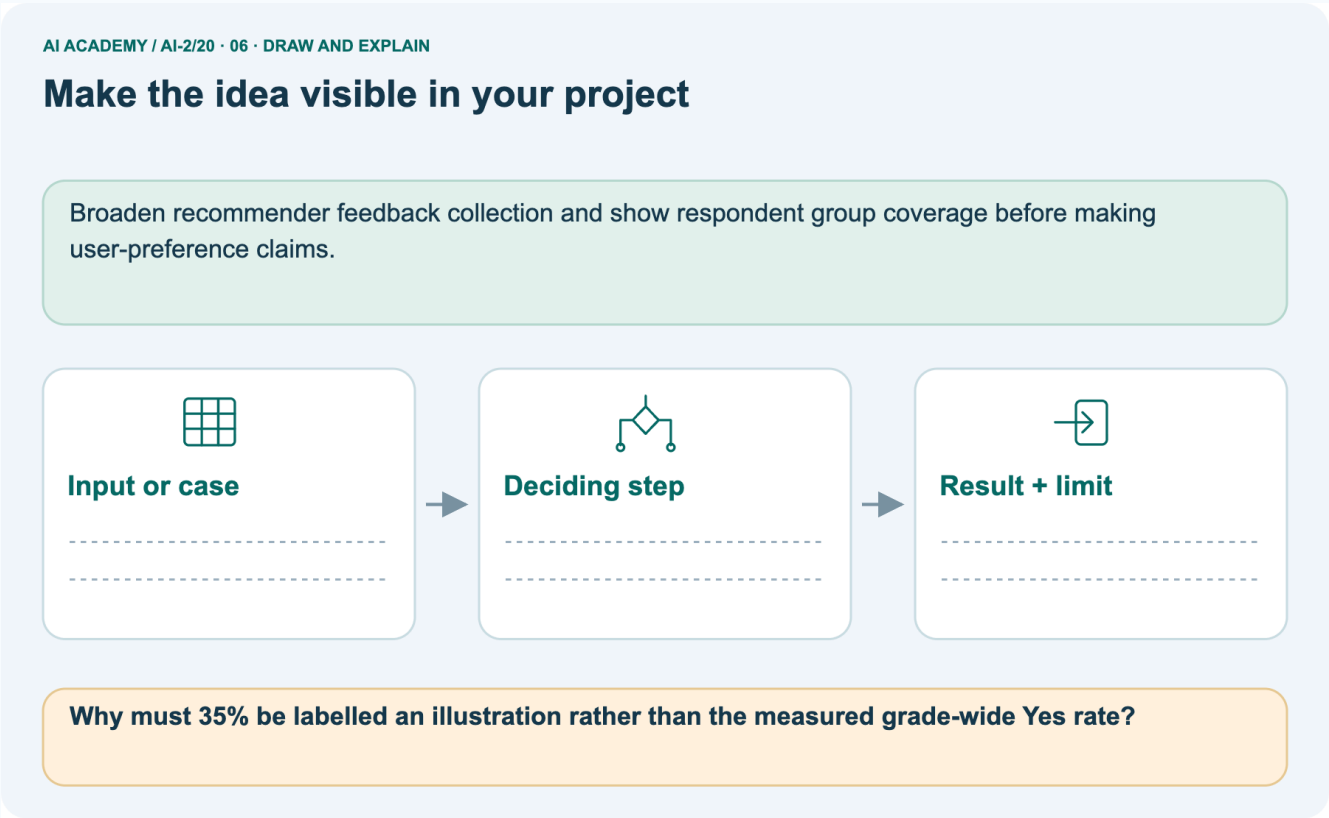
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Neither raw nor weighted volunteer rates prove the whole grade’s opinion; privacy and consent rules still apply.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims. Draw your **input**, deciding step, result and limit.

Supported question: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: It assumes each volunteer group rate represents its whole group; that assumption has not been established.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.